

C++ Algorithms Quiz - 30 Questions with ChatGPT Explanations

Question 1: GCD Program Analysis

```
cpp

int main() {
    int x, y, m, n;
    cin >> x >> y;
    /* x > 0 and y > 0 */
    m = x; n = y;
    while (m != n) {
        if (m > n)
            m = m - n;
        else
            n = n - m;
    }
    cout << n;
}
```

What does the program compute?

- $x + y$ using repeated subtraction
- $x \bmod y$ using repeated subtraction
- ✓ the greatest common divisor of x and y
- the least common multiple of x and y

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 2: Maximum Subarray Sum

Maximum Subarray Sum problem is to find the subarray with maximum sum. For example, given an array (12, -13, 5, 25, -20, 30, 10), the maximum subarray sum is 45.

What will be the worst case time complexity using Divide and Conquer?

- $O(n)$
- ✓ $O(n \log n)$
- $O(\log n)$
- $O(n^2)$

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 3: Power Function Complexity

Consider a situation where you don't have function to calculate power (pow() function in C++) and you need to calculate x^n where x can be any number and n is a positive integer.

What can be the best possible time complexity of your power function?

- $O(n)$
- $O(n \log n)$
- ✓ $O(\log n)$
- $O(n^2)$

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 4: Kth Smallest Element

```
cpp
int partition(int a[], int n);

int kth_smallest(int a[], int n, int k) {
    int left_end = partition(a,n);
    if(left_end == k-1) return a[left_end];
    if(left_end > k-1) return kth_smallest(_____);
    else return kth_smallest(_____);
}
```

Complete the recursive calls:

- ✓ $(a, \text{left_end}, k)$ and $(a+\text{left_end}+1, n-\text{left_end}-1, k-\text{left_end}-1)$
- $(a, \text{left_end}, k)$ and $(a, n-\text{left_end}-1, k-\text{left_end}-1)$
- $(a, \text{left_end}+1, N-\text{left_end}-1, K-\text{left_end}-1)$ and $(a, \text{left_end}, k)$
- None

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 5: Recurrence Relation $T(n) = T(n-1) + n$

The solution is:

- $O(n)$
- $O(n \log n)$
- $O(\log n)$
- ✓ $O(n^2)$

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 6: Recurrence Relation $T(n) = T(n/2) + 1$

The solution is:

- $O(n)$
- $O(n \log n)$
- ✓ $O(\log n)$
- $O(n^2)$

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 7: Recurrence Relation $T(n) = T(n/2) + n$

The solution is:

- ✓ $O(n)$
- $O(n \log n)$
- $O(\log n)$
- $O(n^2)$

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 8: Recurrence Relation $T(n) = 2T(n/2 + 17) + n$

The solution is:

- $O(n)$
- ✓ $O(n \log n)$
- $O(\log n)$
- $O(n^2)$

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 9: Backtracking Algorithm

What happens when the backtracking algorithm reaches a complete solution?

- It backtracks to the root
- ✓ It continues searching for other possible solutions
- It traverses from a different route
- Recursively traverses through the same route

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 10: N-Queens Problem (3x3)

How many fundamental solutions are there for 3 queens on a 3x3 board?

- 1
- 12
- 3
- ✓ 0

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 11: Six Queen vs Five Queen Puzzle

The six queen puzzle has fewer solutions than the five queen puzzle.

- ✓ True
- False
- Can't Determined
- None of these

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 12: Eight Queen Puzzle

How many fundamental solutions are there for the eight queen puzzle?

- 92
- 10
- 11
- ✓ 12

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 13: Merge Sort Code

Choose the correct code for merge sort:

```
cpp

void merge_sort(int arr[], int left, int right) {
    if (left < right) {
        int mid = left+(right-left)/2;
        merge_sort(arr, left, mid);
        merge_sort(arr, mid+1, right);
        merge(arr, left, mid, right);
    }
}
```

✓ This is the correct implementation

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 14: Merge Sort Time Complexity

What is the worst case, best case and average case time complexity of merge sort?

- $O(n^2)$, $O(n)$, $O(n \log n)$
- ✓ $O(n \log n)$, $O(n \log n)$, $O(n \log n)$
- $O(n^2)$, $O(n)$, $O(n)$
- $O(n^2)$, $O(n \log n)$, $O(n \log n)$

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 15: Quick Sort Time Complexity

What is the worst case, best case and average case time complexity of quick sort?

- $O(n^2)$, $O(n)$, $O(n \log n)$
- $O(n \log n)$, $O(n \log n)$, $O(n \log n)$
- $O(n^2)$, $O(n)$, $O(n)$
- ✓ $O(n^2)$, $O(n \log n)$, $O(n \log n)$

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 16: Quick Sort First Phase

Apply Quick sort on: 7 11 14 6 9 4 3 12. What is the sequence after first phase?

- ✓ 6 4 3 7 11 9 14 12
- 6 3 4 7 9 14 11 12
- 7 6 14 11 9 4 3 12
- 7 6 4 3 9 14 11 12

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 17: Quick Sort Maximum Comparisons

Which input sequence will require maximum comparisons?

- ✓ 22 25 56 67 89
- 52 25 76 67 89
- 22 25 76 67 50
- 52 25 89 67 76

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 18: Quick Sort Comparisons

Let t_1 = comparisons for (1, 2, 3, 4, 5) and t_2 = comparisons for (4, 1, 5, 3, 2). Which holds?

- $t_1 = 5$
- $t_1 < t_2$
- ✓ $t_1 > t_2$
- $t_1 = t_2$

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 19: Algorithm Recurrence Matching

Match:

- P. Binary search $\rightarrow$ IV. $T(n) = T(n/2) + 1$
- Q. Merge sort $\rightarrow$ III. $T(n) = 2T(n/2) + cn$
- R. Quick sort $\rightarrow$ I. $T(n) = T(n-k) + T(k) + cn$
- S. Tower of Hanoi $\rightarrow$ II. $T(n) = 2T(n-1) + 1$

✓ P-IV, Q-III, R-I, S-II

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 20: Best Case Time Complexity Order

Correct order of algorithms by best case time complexity:

- ✓ Insertion sort < Quick sort < Merge sort < Selection sort

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 21: Minimum Running Time

Which sorting algorithm has minimum running time in best and average case?

- ✓ Quick sort, Quick sort

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 22: Merge Sort Recurrence

What is the recurrence relation for merge sort?

- ✓ $T(n) = 2T(n/2) + O(n)$

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 23: Merge Sort Space Complexity

The space complexity for merge sort is:

- ✓ $O(n)$

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 24: Min/Max Finding

Minimum number of comparisons to find min and max of 202 integers:

- ✓ 301

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 25: Sorting by Birthdate

True or False: Can sort n people by birthdate (0-2003 A.D.) in $O(n \log n)$ time?

- ✓ False

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 26: Median of Two Sorted Arrays

Optimized worst case time complexity to find median using binary search:

- ✓ $O(\min(\log m, \log n))$

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 27: Polynomial Evaluation

Minimum multiplications for $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$:

- ✓ 3

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 28: Binary Search Time Complexity

Time complexity of binary search:

- ✓ $O(\log n)$

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 29: Quick Sort Space Complexity

Space complexity of quick sort:

- ✓ $O(\log n)$

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Question 30: Permutation Code Output

```
cpp

void permute(string& a, int l, int r) {
    if (l == r)
        cout << a << endl;
    else {
        for (int i = l; i <= r; i++) {
            swap(a[l], a[i]);
            permute(a, l+1, r);
            swap(a[l], a[i]);
        }
    }
}

int main() {
    string a = "ab";
    permute(a, 0, 1);
}
```

Output:

- ✓ ab, ba

ChatGPT's Explanation: *Please provide ChatGPT's explanation for this question*

Notes

To complete this document, please paste ChatGPT's explanations for each question where indicated.